

Multilayer CPA for Cylinders and Spheres

TM ($H_r = 0$) modes in spheres

We consider the multilayer generalization of the notes:

<https://hackmd.io/@aligho/SymTGLcPxg> (<https://hackmd.io/@aligho/SymTGLcPxg>). We first consider the case of no coatings and we generalize to the coated case later in this set of notes.

Without coatings and with layers indexed from $n = 0, 1, 2 \dots, N$ (where each layer covers the region $r_n < r < r_{n+1}$ (with $r_0 \equiv 0$), with corresponding index, n_n , for **TM modes**, we have for the magnetic field:

$$\mathbf{H}_n(r, \theta, \phi) = [a_n h_l^{(1)}(n_n k r) + b_n h_l^{(2)}(n_n k r)] \mathbf{L} Y_{lm}(\theta, \phi),$$

where $k \equiv \omega/c$. And we have the boundary condition (continuity of tangential part of magnetic field since there is no surface current):

$$a_n h_l^{(1)}(n_n k r_{n+1}) + b_n h_l^{(2)}(n_n k r_{n+1}) = a_{n+1} h_l^{(1)}(n_{n+1} k r_{n+1}) + b_{n+1} h_l^{(2)}(n_{n+1} k r_{n+1})$$

We define $x_n \equiv n_n k r_n$ and $y_n \equiv n_n k r_{n+1}$ in order to rewrite this boundary condition as follows:

$$a_n h_l^{(1)}(y_n) + b_n h_l^{(2)}(y_n) = a_{n+1} h_l^{(1)}(x_{n+1}) + b_{n+1} h_l^{(2)}(x_{n+1})$$

Our second boundary condition (continuity of the tangential component of the electric field since there is no surface dipole density) gives us:

$$\varepsilon_{n+1} \left[a_n (y_n h_l^{(1)}(y_n))' + b_n (y_n h_l^{(2)}(y_n))' \right] = \varepsilon_n \left[a_{n+1} (x_{n+1} h_l^{(1)}(x_{n+1}))' + b_{n+1} (x_{n+1} h_l^{(2)}(x_{n+1}))' \right]$$

We write the above in matrix form as follows:

$$\begin{bmatrix} h_l^{(1)}(y_n) & h_l^{(2)}(y_n) \\ \varepsilon_{n+1}(y_n h_l^{(1)}(y_n))' & \varepsilon_{n+1}(y_n h_l^{(2)}(y_n))' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} h_l^{(1)}(x_{n+1}) & h_l^{(2)}(x_{n+1}) \\ \varepsilon_n(x_{n+1} h_l^{(1)}(x_{n+1}))' & \varepsilon_n(x_{n+1} h_l^{(2)}(x_{n+1}))' \end{bmatrix} \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

Therefore, by inverting this matrix relation, we obtain the following relation between the amplitudes in the n 'th and $n + 1$ 'th layers:

$$\begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix} = \frac{1}{\varepsilon_n \left[h_l^{(1)}(x_{n+1})(x_{n+1} h_l^{(2)}(x_{n+1}))' - h_l^{(2)}(x_{n+1})(x_{n+1} h_l^{(1)}(x_{n+1}))' \right]} \times \begin{bmatrix} \varepsilon_n(x_{n+1} h_l^{(2)}(x_{n+1}))' & -h_l^{(2)}(x_{n+1}) \\ -\varepsilon_n(x_{n+1} h_l^{(1)}(x_{n+1}))' & h_l^{(1)}(x_{n+1}) \end{bmatrix} \times \begin{bmatrix} h_l^{(1)}(y_n) & h_l^{(2)}(y_n) \\ \varepsilon_{n+1}(y_n h_l^{(1)}(y_n))' & \varepsilon_{n+1}(y_n h_l^{(2)}(y_n))' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix}$$

Note that we only care about the ratio of a_N/b_N (the ratio of outgoing to incoming wave amplitudes in the environment in which the entire scatterer is placed), so we may rid ourselves of the annoying denominator, giving us:

$$\begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} \varepsilon_n(x_{n+1} h_l^{(2)}(x_{n+1}))' & -h_l^{(2)}(x_{n+1}) \\ -\varepsilon_n(x_{n+1} h_l^{(1)}(x_{n+1}))' & h_l^{(1)}(x_{n+1}) \end{bmatrix} \begin{bmatrix} h_l^{(1)}(y_n) & h_l^{(2)}(y_n) \\ \varepsilon_{n+1}(y_n h_l^{(1)}(y_n))' & \varepsilon_{n+1}(y_n h_l^{(2)}(y_n))' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix}$$

Sanity check

Let's check that this works for $N = 1$. The field in the interior of the sphere has to be a spherical Bessel function of the first kind, which means $a_0 = b_0 = 1$ so we obtain:

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} \varepsilon_0(x_1 h_l^{(2)}(x_1))' & -h_l^{(2)}(x_1) \\ -\varepsilon_0(x_1 h_l^{(1)}(x_1))' & h_l^{(1)}(x_1) \end{bmatrix} \begin{bmatrix} j_l(y_0) \\ \varepsilon_1(y_0 j_l(y_0))' \end{bmatrix}$$

This gives us:

$$\frac{a_1}{b_1} = \frac{\varepsilon_0(x_1 h_l^{(2)}(x_1))' j_l(y_0) - \varepsilon_1(y_0 j_l(y_0))' h_l^{(2)}(x_1)}{-\varepsilon_0(x_1 h_l^{(1)}(x_1))' j_l(y_0) + \varepsilon_1(y_0 j_l(y_0))' h_l^{(1)}(x_1)},$$

as expected.

Second sanity check

As a second sanity check, we consider an $N = 2$ layer case, with a two dimensional coating, where the inner core has dielectric constant ε_0 , the environment has a dielectric constant ε_2 and the two dimensional coating has a thickness dependent dielectric constant $\varepsilon_1 = 1 + i\sigma(\omega)/(d\omega\tilde{\varepsilon}_0)$ (and we take the limit $d \rightarrow 0$ at the end of the calculation). We expect to obtain, after taking the limit, the TM reflection coefficient cited in Localized plasmons in graphene-coated nanospheres

(<https://link.aps.org/pdf/10.1103/PhysRevB.91.125414>) (with the replacement of $j_l(x_2) \rightarrow h_l^{(2)}(x_2)$ in the numerator). Note that we introduce the somewhat unconventional notation $\tilde{\varepsilon}_0$ as the vacuum permittivity. We do this to avoid confusion with the previously introduced relative permittivity of the bulk, ε_0 .

With the definitions of x_i and y_i from the preceding sections, we have, therefore:

$y_0 = \sqrt{\varepsilon_0}(\omega/c)a$, $x_1 = \sqrt{\varepsilon_1}(\omega/c)a$, $y_1 = \sqrt{\varepsilon_1}(\omega/c)(a + d)$ and $x_2 = \sqrt{\varepsilon_2}(\omega/c)(a + d)$, where we have introduced the bulk sphere radius, a . In the following derivation, we may (or may not) use the notation $n_i \equiv \sqrt{\varepsilon_i}$, for convenience.

Our expression for the amplitudes in region two (the environment) are given by:

$$\begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} \varepsilon_1(x_2 h_l^{(2)}(x_2))' & -h_l^{(2)}(x_2) \\ -\varepsilon_1(x_2 h_l^{(1)}(x_2))' & h_l^{(1)}(x_2) \end{bmatrix} \times \begin{bmatrix} h_l^{(1)}(y_1) & h_l^{(2)}(y_1) \\ \varepsilon_2(y_1 h_l^{(1)}(y_1))' & \varepsilon_2(y_1 h_l^{(2)}(y_1))' \end{bmatrix} \times \begin{bmatrix} \varepsilon_0(x_1 h_l^{(2)}(x_1))' & -h_l^{(2)}(x_1) \\ -\varepsilon_0(x_1 h_l^{(1)}(x_1))' & h_l^{(1)}(x_1) \end{bmatrix} \begin{bmatrix} j_l(y_0) \\ \varepsilon_1(y_0 j_l(y_0))' \end{bmatrix}$$

We multiply the penultimate matrix by the last matrix to give us:

$$\begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} \varepsilon_1(x_2 h_l^{(2)}(x_2))' & -h_l^{(2)}(x_2) \\ -\varepsilon_1(x_2 h_l^{(1)}(x_2))' & h_l^{(1)}(x_2) \end{bmatrix} \times \begin{bmatrix} h_l^{(1)}(y_1) & h_l^{(2)}(y_1) \\ \varepsilon_2(y_1 h_l^{(1)}(y_1))' & \varepsilon_2(y_1 h_l^{(2)}(y_1))' \end{bmatrix} \times \begin{bmatrix} \varepsilon_0(x_1 h_l^{(2)}(x_1))' j_l(y_0) - \varepsilon_1 h_l^{(2)}(x_1) (y_0 j_l(y_0))' \\ -\varepsilon_0(x_1 h_l^{(1)}(x_1))' j_l(y_0) + \varepsilon_1 h_l^{(1)}(x_1) (y_0 j_l(y_0))' \end{bmatrix},$$

Note that the last element of this expression is a 2×1 vector not a 2×2 matrix. We next multiply the first two matrices to give us:

$$\begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} \varepsilon_1(x_2 h_l^{(2)}(x_2))' h_l^{(1)}(y_1) - \varepsilon_2(y_1 h_l^{(1)}(y_1))' h_l^{(2)}(x_2) & \varepsilon_1(x_2 h_l^{(2)}(x_2))' h_l^{(2)}(y_1) - \varepsilon_2 h_l^{(2)}(x_2) (y_1 h_l^{(2)}(y_1))' \\ \varepsilon_2(y_1 h_l^{(1)}(y_1))' h_l^{(1)}(x_2) - \varepsilon_1(x_2 h_l^{(1)}(x_2))' h_l^{(1)}(y_1) & \varepsilon_2(y_1 h_l^{(2)}(y_1))' h_l^{(1)}(x_2) - \varepsilon_1 h_l^{(2)}(y_1) (x_2 h_l^{(1)}(x_2))' \end{bmatrix} \times \begin{bmatrix} \varepsilon_0(x_1 h_l^{(2)}(x_1))' j_l(y_0) - \varepsilon_1 h_l^{(2)}(x_1) (y_0 j_l(y_0))' \\ -\varepsilon_0(x_1 h_l^{(1)}(x_1))' j_l(y_0) + \varepsilon_1 h_l^{(1)}(x_1) (y_0 j_l(y_0))' \end{bmatrix}$$

We now expand each of a_2, b_2 into terms proportional to $\varepsilon_1^2, \varepsilon_1 \varepsilon_0, 2$ and $\varepsilon_2 \varepsilon_0$. We denote these by superscripts, i.e. $a_2 = a_2^{11} + a_2^{10} + a_2^{12} + a_2^{02}$. Let's first restrict ourselves to a_2 and consider the terms proportional to $\varepsilon_2 \varepsilon_0$:

$$a_2^{02} = \varepsilon_0 \varepsilon_2 j_l(y_0) h_l^{(2)}(x_2) \left[(x_1 h_l^{(1)}(x_1))' (y_1 h_l^{(2)}(y_1))' - (x_1 h_l^{(2)}(x_1))' (y_1 h_l^{(1)}(y_1))' \right] \approx \varepsilon_0 \varepsilon_2 j_l(y_0) h_l^{(2)}(x_2) \left[e^{i(x_1 - y_1)} - e^{-i(x_1 - y_1)} \right] \approx -2i \varepsilon_0 \varepsilon_2 j_l(y_0) h_l^{(2)}(x_2) \sqrt{\varepsilon_1} (\omega/c) d \rightarrow 0,$$

where we used the fact that $|x_1|, |y_1| \gg 1$ (in the limit $d \rightarrow 0$) in order to use asymptotic expansions of the spherical Hankel functions. Note however, that the difference, $y_1 - x_1 = \sqrt{\varepsilon_1} (d\omega/c) \rightarrow 0$ as $d \rightarrow 0$. In particular, a_2^{02} scales as $d^{1/2}$ as $d \rightarrow 0$. We now move on to a_2^{11} , which reads as follows:

$$\begin{aligned} a_2^{11} &= \varepsilon_1^2 (y_0 j_l(y_0))' (x_2 h_l^{(2)}(x_2))' [h_l^{(1)}(x_1) h_l^{(2)}(y_1) - h_l^{(2)}(x_1) h_l^{(1)}(y_1)] \approx \\ &\quad \varepsilon_1^2 (y_0 j_l(y_0))' (x_2 h_l^{(2)}(x_2))' \frac{1}{x_1 y_1} \left[e^{i(x_1 - y_1)} - e^{-i(x_1 - y_1)} \right] = \\ &\quad -i \varepsilon_1 (y_0 j_l(y_0))' (x_2 h_l^{(2)}(x_2))' \frac{2\sqrt{\varepsilon_1} (\omega/c) d}{(\omega/c)^2 [a^2 + ad]} \\ &\approx -\frac{2i\sqrt{\varepsilon_1}}{(\omega/c)a} (y_0 j_l(y_0))' (x_2 h_l^{(2)}(x_2))' (i\sigma(\omega)/(a\omega\varepsilon_0)) \end{aligned}$$

Note that this scales as $d^{-1/2}$ as $d \rightarrow 0$ (which reinforces our justification for ignoring the contribution of a_2^{11} above. Now we investigate the remaining terms, i.e. a_2^{01} and a_2^{12} :

$$a_2^{10} = \varepsilon_0 \varepsilon_1 j_l(y_0) (x_2 h_l^{(2)}(x_2))' \left[h_l^{(1)}(y_1) (x_1 h_l^{(2)}(x_1))' - h_l^{(2)}(y_1) (x_1 h_l^{(1)}(x_1))' \right] \approx$$

$$-2i\varepsilon_0 \sqrt{\varepsilon_1} j_l(y_0) (x_2 h_l^{(2)}(x_2))' \frac{1}{(\omega/c)a}$$

$$a_2^{12} = \varepsilon_2 \varepsilon_1 h_l^{(2)}(x_2) (y_0 j_l(y_0))' [h_l^{(2)}(x_1) (y_1 h_l^{(1)}(y_1))' - h_l^{(1)}(x_1) (y_1 h_l^{(2)}(y_1))'] \approx$$

$$2i\varepsilon_2 \sqrt{\varepsilon_1} h_l^{(2)}(x_2) (y_0 j_l(y_0))' \frac{1}{(\omega/c)a}$$

Therefore, we may write:

$$a_2^{11} + a_2^{10} + a_2^{12} \approx \frac{2i\sqrt{\varepsilon_1}}{a\omega/c} \left[-\varepsilon_0 j_l(y_0) (x_2 h_l^{(2)}(x_1))' - \right.$$

$$\left. (i\sigma(\omega)/\omega \tilde{\varepsilon}_0) (y_0 j_l(y_0))' (x_2 h_l^{(2)}(x_2))' + \varepsilon_2 h_l^{(2)}(x_2) (y_0 j_l(y_0))' \right]$$

Note that we do not have to independently calculate b_2 . We can simply note that the matrix elements involved in b_2 are the same as those for a_2 with the simple replacement $h_l^{(2)}(x_2) \rightarrow -h_l^{(1)}(x_2)$. Therefore, we get:

$$\frac{a_2}{b_2} = \frac{-\varepsilon_0 j_l(y_0) (x_2 h_l^{(2)}(x_1))' - (i\sigma(\omega)/\omega \tilde{\varepsilon}_0) (y_0 j_l(y_0))' (x_2 h_l^{(2)}(x_2))' + \varepsilon_2 h_l^{(2)}(x_2) (y_0 j_l(y_0))'}{\varepsilon_0 j_l(y_0) (x_2 h_l^{(1)}(x_1))' + (i\sigma(\omega)/\omega \tilde{\varepsilon}_0) (y_0 j_l(y_0))' (x_2 h_l^{(1)}(x_2))' - \varepsilon_2 h_l^{(1)}(x_2) (y_0 j_l(y_0))'},$$

which is exactly the TM reflection coefficient cited in Localized plasmons in graphene-coated nanospheres (<https://link.aps.org/pdf/10.1103/PhysRevB.91.125414>) (with the replacement of $j_l(x_2) \rightarrow h_l^{(2)}(x_2)$ in the numerator).

For TE ($E_r = 0$) modes

Referring to https://hackmd.io/@aligho/ByRI-iJ_ge (https://hackmd.io/@aligho/ByRI-iJ_ge),

$$\mathbf{E}_n(r, \theta, \phi) = [a_n h_l^{(1)}(n_n k r) + b_n h_l^{(2)}(n_n k r)] \mathbf{L} Y_{lm}(\theta, \phi)$$

And we have the boundary condition on continuity of the tangential component of the electric field, which reads:

$$a_n h_l^{(1)}(n_n k r_{n+1}) + b_n h_l^{(2)}(n_n k r_{n+1}) = a_{n+1} h_l^{(1)}(n_{n+1} k r_{n+1}) + b_{n+1} h_l^{(2)}(n_{n+1} k r_{n+1})$$

We define $x_n \equiv n_n k r_n$ and $y_n \equiv n_n k r_{n+1}$ to write this as well as the boundary condition on the tangential component of the magnetic field as follows:

$$a_n h_l^{(1)}(y_n) + b_n h_l^{(2)}(y_n) = a_{n+1} h_l^{(1)}(x_{n+1}) + b_{n+1} h_l^{(2)}(x_{n+1})$$

$$\left[a_n (y_n h_l^{(1)}(y_n))' + b_n (y_n h_l^{(2)}(y_n))' \right] = \left[a_{n+1} (x_{n+1} h_l^{(1)}(x_{n+1}))' + b_{n+1} (x_{n+1} h_l^{(2)}(x_{n+1}))' \right]$$

This gives (modulo an unimportant overall factor):

$$\begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} (x_{n+1} h_l^{(2)}(x_{n+1}))' & -h_l^{(2)}(x_{n+1}) \\ -(x_{n+1} h_l^{(1)}(x_{n+1}))' & h_l^{(1)}(x_{n+1}) \end{bmatrix} \begin{bmatrix} h_l^{(1)}(y_n) & h_l^{(2)}(y_n) \\ (y_n h_l^{(1)}(y_n))' & (y_n h_l^{(2)}(y_n))' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix}$$

Sanity check

For a one-layer system, taking $a_0 = b_0$, we obtain:

$$\frac{a_1}{b_1} = \frac{j_l(y_0)(x_1 h_l^{(2)}(x_1))' - (y_0 j_l(y_0))' h_l^{(2)}(x_1)}{-j_l(y_0)(x_1 h_l^{(1)}(x_1))' + (y_0 j_l(y_0))' h_l^{(1)}(x_1)}$$

For electric ($H = H_z$) modes in cylinders

We again consider the multilayer generalization of these notes

<https://hackmd.io/@aligho/SymTGLcPxg> (<https://hackmd.io/@aligho/SymTGLcPxg>) but for cylinders this time. We again take the layers to be indexed from $0, 1, 2, \dots, N$, with each layer corresponding to the region $r_n < r < r_{n+1}$ and we define $x_n \equiv n_n k r_n$, $y_n \equiv n_n k r_{n+1}$, with $k \equiv \omega/c$.

$$\text{We take } \mathbf{H}_n(r, \theta) = e^{im\phi} \left[a_n H_m^{(1)}(n_n k r) + b_n H_m^{(2)}(n_n k r) \right] \mathbf{e}_z$$

As before, we require $a_0 = b_0$ so that the field does not diverge at the origin (i.e. so we have no nonzero coefficient of the Bessel function of the second kind at the origin). In addition, our boundary conditions give us the following relations:

$$a_n H_m^{(1)}(y_n) + b_n H_m^{(2)}(y_n) = a_{n+1} H_m^{(1)}(x_{n+1}) + b_{n+1} H_m^{(2)}(x_{n+1})$$

$$n_{n+1} \begin{bmatrix} a_n H_m^{(1)}(y_n)' + b_n H_m^{(2)}(y_n)' \\ n_{n+1} H_m^{(1)}(x_{n+1})' + b_{n+1} H_m^{(2)}(x_{n+1})' \end{bmatrix} = n_n \begin{bmatrix} a_{n+1} H_m^{(1)}(x_{n+1})' + b_{n+1} H_m^{(2)}(x_{n+1})' \\ n_n H_m^{(1)}(y_n)' + b_n H_m^{(2)}(y_n)' \end{bmatrix}$$

As in the case of the sphere modes, we write this in matrix form:

$$\begin{bmatrix} H_m^{(1)}(y_n) & H_m^{(2)}(y_n) \\ n_{n+1} H_m^{(1)}(y_n)' & n_{n+1} H_m^{(2)}(y_n)' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} H_m^{(1)}(x_{n+1}) & H_m^{(2)}(x_{n+1}) \\ n_n H_m^{(1)}(x_{n+1})' & n_n H_m^{(2)}(x_{n+1})' \end{bmatrix} \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

We invert this relationship (note that, as before, the determinant is irrelevant if we only care about the ratio of the fields). Therefore, we obtain the following:

$$\begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} n_n H_m^{(2)}(x_{n+1})' & -H_m^{(2)}(x_{n+1}) \\ -n_n H_m^{(1)}(x_{n+1})' & H_m^{(1)}(x_{n+1}) \end{bmatrix} \begin{bmatrix} H_m^{(1)}(y_n) & H_m^{(2)}(y_n) \\ n_{n+1} H_m^{(1)}(y_n)' & n_{n+1} H_m^{(2)}(y_n)' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix}$$

Sanity check

For a single layer, we have the following:

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} n_0 H_m^{(2)}(x_1)' & -H_m^{(2)}(x_1) \\ -n_0 H_m^{(1)}(x_1)' & H_m^{(1)}(x_1) \end{bmatrix} \begin{bmatrix} J_m(y_0) \\ n_1 J_m'(y_0) \end{bmatrix},$$

which gives us the ratio:

$$\frac{a_1}{b_1} = \frac{n_0 H_m^{(2)}(x_1)' J_m(y_0) - n_1 J_m'(y_0) H_m^{(2)}(x_1)}{n_1 J_m'(y_0) H_m^{(1)}(x_1) - n_0 J_m(y_0) H_m^{(1)}(x_1)'}$$

Second sanity check

As with the second sanity check we did for the modes of a sphere, we examine the case of a $N = 2$ layer cylinder, with

$$n_1^2 = 1 + \frac{i\sigma(\omega)}{\tilde{\epsilon}_0 \omega d}$$

and

$$y_0 = n_0(\omega/c)a, x_1 = n_1(\omega/c)a, y_1 = n_1(\omega/c)(a+d), x_2 = n_2(\omega/c)(a+d),$$

where we are concerned with the $d \rightarrow 0$ limit (with $\sigma(\omega)$ remaining fixed).

$$\begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} n_1 H_m^{(2)}(x_2)' & -H_m^{(2)}(x_2) \\ -n_1 H_m^{(1)}(x_2)' & H_m^{(1)}(x_2) \end{bmatrix} \begin{bmatrix} H_m^{(1)}(y_1) & H_m^{(2)}(y_1) \\ n_2 H_m^{(1)}(y_1)' & n_2 H_m^{(2)}(y_1)' \end{bmatrix} \times \\ \begin{bmatrix} n_0 H_m^{(2)}(x_1)' J_m(y_0) - n_1 J_m'(y_0) H_m^{(2)}(x_1) \\ n_1 J_m'(y_0) H_m^{(1)}(x_1) - n_0 J_m(y_0) H_m^{(1)}(x_1)' \end{bmatrix},$$

where the last element is a 2×1 vector, not a 2×2 matrix. We multiply the first two matrices to give us:

$$\begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} n_1 H_m^{(2)}(x_2)' H_m^{(1)}(y_1) - n_2 H_m^{(1)}(y_1)' H_m^{(2)}(x_2) & n_1 H_m^{(2)}(x_2)' H_m^{(2)}(y_1) - n_2 H_m^{(2)}(y_1)' H_m^{(2)}(x_2) \\ n_2 H_m^{(1)}(y_1)' H_m^{(1)}(x_2) - n_1 H_m^{(1)}(x_2)' H_m^{(1)}(y_1) & n_2 H_m^{(2)}(y_1)' H_m^{(1)}(x_2) - n_1 H_m^{(1)}(x_2)' H_m^{(2)}(y_1) \end{bmatrix} \times \\ \begin{bmatrix} n_0 H_m^{(2)}(x_1)' J_m(y_0) - n_1 J_m'(y_0) H_m^{(2)}(x_1) \\ n_1 J_m'(y_0) H_m^{(1)}(x_1) - n_0 J_m(y_0) H_m^{(1)}(x_1)' \end{bmatrix}$$

As in the case of the sphere, we do not need to solve for b_2 if we solve for a_2 . We can simply make the replacement $H_m^{(2)}(x_2) \rightarrow -H_m^{(1)}(x_2)$. Therefore, we just solve for a_2 . We separate the terms into those proportional to $n_0 n_1$, $n_0 n_2$, n_1^2 , and $n_1 n_2$. We start with the term proportional to $n_1 n_2$:

$$n_1 n_2 H_m^{(2)}(x_2) J_m'(y_0) \left[H_m^{(1)}(y_1)' H_m^{(2)}(x_1) - H_m^{(2)}(y_1)' H_m^{(1)}(x_1) \right] \approx \frac{2i n_1 n_2 H_m^{(2)}(x_2) J_m'(y_0)}{n_1 (\omega/c) a} \frac{2}{\pi}$$

The term proportional to $n_0 n_1$ reads:

$$n_0 n_1 H_m^{(2)}(x_2)' J_m(y_0) \left[H_m^{(2)}(x_1)' H_m^{(1)}(y_1) - H_m^{(1)}(x_1)' H_m^{(2)}(y_1) \right] \approx -\frac{2i n_0 n_1 H_m^{(2)}(x_2)' J_m(y_0)}{n_1 (\omega/c) a} \frac{2}{\pi}$$

The term proportional to n_1^2 reads:

$$n_1^2 J_m'(y_0) H_m^{(2)}(x_2)' \left[H_m^{(1)}(x_1) H_m^{(2)}(y_1) - H_m^{(2)}(x_1) H_m^{(1)}(y_1) \right] \approx -\frac{2i n_1^3 (\omega/c) d}{n_1 (\omega/c) a} \frac{2}{\pi}$$

The term proportional to $n_0 n_2$ reads:

$$n_0 n_2 J_m(y_0) H_m^{(2)}(x_2) \left[H_m^{(1)}(x_1)' H_m^{(2)}(y_1)' - H_m^{(1)}(y_1)' H_m^{(2)}(x_1)' \right] \approx 0$$

Therefore, we obtain:

$$a_2 \approx n_2 H_m^{(2)}(x_2) J'_m(y_0) - n_0 H_m^{(2)}(x_2)' J_m(y_0) - \frac{i\sigma(\omega)}{\tilde{\epsilon}_0 \omega d} \frac{\omega}{c} d J'_m(y_0) H_m^{(2)}(x_2)' =$$

$$n_2 H_m^{(2)}(x_2) J'_m(y_0) - n_0 H_m^{(2)}(x_2)' J_m(y_0) - \frac{i\sigma(\omega)}{c \tilde{\epsilon}_0} J'_m(y_0) H_m^{(2)}(x_2)'$$

So the scattering coefficient is given by:

$$\frac{a_2}{b_2} = \frac{n_2 H_m^{(2)}(x_2) J'_m(y_0) - n_0 H_m^{(2)}(x_2)' J_m(y_0) - \frac{i\sigma(\omega)}{c \tilde{\epsilon}_0} J'_m(y_0) H_m^{(2)}(x_2)'}{-n_2 H_m^{(1)}(x_2) J'_m(y_0) + n_0 H_m^{(1)}(x_2)' J_m(y_0) + \frac{i\sigma(\omega)}{c \tilde{\epsilon}_0} J'_m(y_0) H_m^{(1)}(x_2)'},$$

which is the same as the result derived here: <https://hackmd.io/@aligho/B1Gs5Cz2el>
(<https://hackmd.io/@aligho/B1Gs5Cz2el>).

For magnetic ($E = E_z$) modes in cylinders

We again consider the multilayer generalization of these notes

<https://hackmd.io/@aligho/SymTGLcPxg> (<https://hackmd.io/@aligho/SymTGLcPxg>) but for $E = E_z$ modes in cylinders. We again take the layers to be indexed from $n = 0, 1, 2, \dots, N$, with each layer corresponding to the region $r_n < r < r_{n+1}$ (again taking $r_0 = 0$) and we define $x_n \equiv n_n k r_n$, $y_n \equiv n_n k r_{n+1}$, with $k \equiv \omega/c$.

$$\text{We take } \mathbf{E}_n(r, \theta) = e^{im\phi} \left[a_n H_m^{(1)}(n_n k r) + b_n H_m^{(2)}(n_n k r) \right] \mathbf{e}_z$$

As before, we require $a_0 = b_0$ so that the field does not diverge at the origin. In addition, our boundary conditions give us the conditions:

$$a_n H_m^{(1)}(y_n) + b_n H_m^{(2)}(y_n) = a_{n+1} H_m^{(1)}(x_{n+1}) + b_{n+1} H_m^{(2)}(x_{n+1})$$

(Continuity of the tangential part of the electric field). And:

$$n_n \left[a_n H_m^{(1)}(y_n)' + b_n H_m^{(2)}(y_n)' \right] = n_{n+1} \left[a_{n+1} H_m^{(1)}(x_{n+1})' + b_{n+1} H_m^{(2)}(x_{n+1})' \right]$$

(Continuity of the tangential component of the magnetic field). As in the case of the sphere modes, we write this in matrix form:

$$\begin{bmatrix} H_m^{(1)}(y_n) & H_m^{(2)}(y_n) \\ n_n H_m^{(1)}(y_n)' & n_n H_m^{(2)}(y_n)' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} H_m^{(1)}(x_{n+1}) & H_m^{(2)}(x_{n+1}) \\ n_{n+1} H_m^{(1)}(x_{n+1})' & n_{n+1} H_m^{(2)}(x_{n+1})' \end{bmatrix} \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

which gives us (up to an unimportant factor):

$$\begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} n_{n+1} H_m^{(2)}(x_{n+1})' & -H_m^{(2)}(x_{n+1}) \\ -n_{n+1} H_m^{(1)}(x_{n+1})' & H_m^{(1)}(x_{n+1}) \end{bmatrix} \begin{bmatrix} H_m^{(1)}(y_n) & H_m^{(2)}(y_n) \\ n_n H_m^{(1)}(y_n)' & n_n H_m^{(2)}(y_n)' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix}$$

Sanity check

For a single layer, taking $a_0 = b_0$, we obtain:

$$\frac{a_1}{b_1} = \frac{n_1 J_m(y_0) H_m^{(2)}(x_1)' - n_0 J_m(y_0)' H_m^{(2)}(x_1)}{-n_1 J_m(y_0) H_m^{(1)}(x_1)' + n_0 J_m(y_0)' H_m^{(1)}(x_1)}$$

Result with a coating for spherical systems (TM modes)

In the following, we generalize the results derived above to account for the presence of a surface current at the boundaries of each set of consecutive layers. There is no substantial conceptual difficulty in doing so. However, the results are a bit messier. In this section, we consider spherical layers and transverse-magnetic modes.

We again take for the magnetic field the following:

$$\mathbf{H}_n(r, \theta, \phi) = [a_n h_l^{(1)}(n_n k r) + b_n h_l^{(2)}(n_n k r)] \mathbf{L} Y_{lm}(\theta, \phi),$$

As in the above, we define $x_n \equiv n_n k r_n$ and $y_n \equiv n_n k r_{n+1}$ in order to write the boundary conditions as follows:

$$\begin{aligned}
& \left[a_{n+1} h_l^{(1)}(x_{n+1}) + b_{n+1} h_l^{(2)}(x_{n+1}) \right] - \left[a_n h_l^{(1)}(y_n) + b_n h_l^{(2)}(y_n) \right] = \\
& \quad \frac{i\sigma_{n,n+1}(\omega)}{\omega\epsilon_0\epsilon_n r_{n+1}} \left[a_n (y_n h_l^{(1)}(y_n))' + b_n (y_n h_l^{(2)}(y_n))' \right] \rightarrow \\
& \left[a_{n+1} h_l^{(1)}(x_{n+1}) + b_{n+1} h_l^{(2)}(x_{n+1}) \right] = a_n \left[h_l^{(1)}(y_n) + \frac{i\sigma_{n,n+1}(\omega)}{\omega\epsilon_0\epsilon_n r_{n+1}} (y_n h_l^{(1)}(y_n))' \right] + \\
& \quad b_n \left[h_l^{(2)}(y_n) + \frac{i\sigma_{n,n+1}(\omega)}{\omega\epsilon_0\epsilon_n r_{n+1}} (y_n h_l^{(2)}(y_n))' \right],
\end{aligned}$$

which gives the change in the tangential component of the magnetic field due to finite surface current (the surface current being sourced by a surface conductivity between layers n and $n + 1$, denoted by $\sigma_{n,n+1}(\omega)$). The surface current changes nothing about the continuity of the tangential component of the electric field, so this still reads, as before, as follows:

$$\epsilon_{n+1} \left[a_n (y_n h_l^{(1)}(y_n))' + b_n (y_n h_l^{(2)}(y_n))' \right] = \epsilon_n \left[a_{n+1} (x_{n+1} h_l^{(1)}(x_{n+1}))' + b_{n+1} (x_{n+1} h_l^{(2)}(x_{n+1}))' \right]$$

We write these in matrix form:

$$\begin{bmatrix} h_l^{(1)}(y_n) + i(y_n h_l^{(1)}(y_n))' \sigma_{n,n+1}(\omega) / (\omega\epsilon_0\epsilon_n r_{n+1}) & h_l^{(2)}(y_n) + i(y_n h_l^{(2)}(y_n))' \sigma_{n,n+1}(\omega) / (\omega\epsilon_0\epsilon_n r_{n+1}) \\ \epsilon_{n+1} (y_n h_l^{(1)}(y_n))' & \epsilon_{n+1} (y_n h_l^{(2)}(y_n))' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} h_l^{(1)}(x_{n+1}) & h_l^{(2)}(x_{n+1}) \\ \epsilon_n (x_{n+1} h_l^{(1)}(x_{n+1}))' & \epsilon_n (x_{n+1} h_l^{(2)}(x_{n+1}))' \end{bmatrix} \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

As before, we invert this relation and neglect an overall coefficient, to give us:

$$\begin{bmatrix} h_l^{(1)}(y_n) + i(y_n h_l^{(1)}(y_n))' \sigma_{n,n+1}(\omega) / (\omega\epsilon_0\epsilon_n r_{n+1}) & h_l^{(2)}(y_n) + i(y_n h_l^{(2)}(y_n))' \sigma_{n,n+1}(\omega) / (\omega\epsilon_0\epsilon_n r_{n+1}) \\ \epsilon_{n+1} (y_n h_l^{(1)}(y_n))' & \epsilon_{n+1} (y_n h_l^{(2)}(y_n))' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} \epsilon_n (x_{n+1} h_l^{(2)}(x_{n+1}))' & -h_l^{(2)}(x_{n+1}) \\ -\epsilon_n (x_{n+1} h_l^{(1)}(x_{n+1}))' & h_l^{(1)}(x_{n+1}) \end{bmatrix} \times \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

Sanity check

If the scatterer only has one layer, we take $a_0 = b_0 = 1$. Furthermore, for convenience, since we have just one interface, we take $\sigma(\omega) \equiv \sigma_{n,n+1}(\omega)$. In addition, we take the permittivity of the sphere's interior to be ε_m and the permittivity of the environment to be ε_d . We thus obtain the following:

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} \varepsilon_m(x_1 h_l^{(2)}(x_1))' & -h_l^{(2)}(x_1) \\ -\varepsilon_m(x_1 h_l^{(1)}(x_1))' & h_l^{(1)}(x_1) \end{bmatrix} \times \begin{bmatrix} j_l^{(1)}(y_0) + i(y_0 j_l^{(1)}(y_0))' \sigma_{n,n+1}(\omega) / (\omega \varepsilon_0 \varepsilon_m r_1) \\ \varepsilon_d(y_0 j_l^{(1)}(y_0))' \end{bmatrix}$$

From this, we may obtain the ratio of outgoing to incoming fields, which will be given by the following:

$$\frac{a_1}{b_1} = \frac{\varepsilon_m j_l(y_0)(x_1 h_l^{(2)}(x_1))' - \varepsilon_d h_l^{(2)}(x_1)(y_0 j_l(y_0))' + i(x_1 h_l^{(2)}(x_1))'(y_0 j_l(y_0))' \sigma(\omega) / (\omega \varepsilon_0 R)}{-\varepsilon_m j_l(y_0)(x_1 h_l^{(1)}(x_1))' + \varepsilon_d h_l^{(1)}(x_1)(y_0 j_l(y_0))' - i(x_1 h_l^{(1)}(x_1))'(y_0 j_l(y_0))' \sigma(\omega) / (\omega \varepsilon_0 R)},$$

where, for convenience, we defined $R \equiv r_1$.

Result with a coating for cylindrical systems ($H = H_z$ modes)

The change in the tangential part of the magnetic field gives us:

$$\begin{aligned} & \left[a_{n+1} H_m^{(1)}(x_{n+1}) + b_{n+1} H_m^{(2)}(x_{n+1}) \right] - \left[a_n H_m^{(1)}(y_n) + b_n H_m^{(2)}(y_n) \right] = \\ & \quad \frac{i\sigma_{n,n+1}(\omega)}{c\varepsilon_0 n_n} \left[a_n H_m^{(1)}(y_n)' + b_n H_m^{(2)}(y_n)' \right] \rightarrow \\ & \left[a_{n+1} H_m^{(1)}(x_{n+1}) + b_{n+1} H_m^{(2)}(x_{n+1}) \right] = a_n \left[H_m^{(1)}(y_n) + \frac{i\sigma_{n,n+1}(\omega)}{c\varepsilon_0 n_n} H_m^{(1)}(y_n)' \right] + \\ & \quad b_n \left[H_m^{(2)}(y_n) + \frac{i\sigma_{n,n+1}(\omega)}{c\varepsilon_0 n_n} H_m^{(2)}(y_n)' \right] (y_n)' \end{aligned}$$

Continuity of the electric field gives us:

$$n_{n+1} \left[a_n H_m^{(1)}(y_n)' + b_n H_m^{(2)}(y_n)' \right] = n_n \left[a_{n+1} H_m^{(1)}(x_{n+1})' + b_{n+1} H_m^{(2)}(x_{n+1})' \right]$$

As in the case of the sphere modes, we write this in matrix form:

$$\begin{bmatrix} H_m^{(1)}(y_n) + i\sigma_{n,n+1}(\omega)H_m^{(1)}(y_n)'/c\varepsilon_0 n_n & H_m^{(2)}(y_n) + i\sigma_{n,n+1}(\omega)H_m^{(2)}(y_n)'/c\varepsilon_0 n_n \\ n_{n+1}H_m^{(1)}(y_n)' & n_{n+1}H_m^{(2)}(y_n)' \end{bmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} H_m^{(1)}(x_{n+1}) & H_m^{(2)}(x_{n+1}) \\ n_n H_m^{(1)}(x_{n+1})' & n_n H_m^{(2)}(x_{n+1})' \end{bmatrix} \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

We invert this relationship (note that, as before, the determinant is irrelevant if we only care about the ratio of the fields). Therefore, we obtain the following:

$$\begin{bmatrix} H_m^{(1)}(y_n) + i\sigma_{n,n+1}(\omega)H_m^{(1)}(y_n)'/c\varepsilon_0 n_n & H_m^{(2)}(y_n) + i\sigma_{n,n+1}(\omega)H_m^{(2)}(y_n)'/c\varepsilon_0 n_n \\ n_{n+1}H_m^{(1)}(y_n)' & n_{n+1}H_m^{(2)}(y_n)' \end{bmatrix} \times \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} n_n H_m^{(2)}(x_{n+1})' & -H_m^{(2)}(x_{n+1}) \\ -n_n H_m^{(1)}(x_{n+1})' & H_m^{(1)}(x_{n+1}) \end{bmatrix} \times \begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix}$$

Sanity check

For a single layer, we have the following:

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} n_0 H_m^{(2)}(x_1)' & -H_m^{(2)}(x_1) \\ -n_0 H_m^{(1)}(x_1)' & H_m^{(1)}(x_1) \end{bmatrix} \begin{bmatrix} J_m(y_0) + i\sigma_{0,1}(\omega)J_m'(y_0)/c\varepsilon_0 n_0 \\ n_1 J_m'(y_0) \end{bmatrix},$$

which gives us the ratio:

$$\frac{a_1}{b_1} = \frac{n_0 H_m^{(2)}(x_1)' J_m(y_0) - n_1 J_m'(y_0) H_m^{(2)}(x_1) + i\sigma_{0,1}(\omega) J_m'(y_0) H_m^{(2)}(x_1)'/c\varepsilon_0}{n_1 J_m'(y_0) H_m^{(1)}(x_1) - n_0 J_m(y_0) H_m^{(1)}(x_1)' - i\sigma_{0,1}(\omega) J_m'(y_0) H_m^{(1)}(x_1)'/c\varepsilon_0},$$

which is consistent with what we derived in this set of notes:

https://hackmd.io/@aligho/B1Gs5Cz2el#With-a-graphene-cloak-HH_z

(https://hackmd.io/@aligho/B1Gs5Cz2el#With-a-graphene-cloak-HH_z).